Dr. Soham Poddar

PhD from Indian Institute of Technology, Kharagpur

PERSONAL DETAILS

Contact	sohampoddar26@gmail.com
Website	https://sohampoddar26.github.io/
Google Scholar	https://scholar.google.com/citations?user=PIIZaa0AAAAJ&hl=en/
Research Interests	Natural Language Processing, Large Language Models, Social Computing, Agentic AI

INDUSTRY EXPERIENCE

Research Scientist 2026-Present
NielsenIQ
Designing and developing Custom Language Models for product-market data.

Research Intern 2025-2026
Hewlett Packard Enterprise Labs
Developing and benchmarking different LLM Agentic frameworks for context engineering, long-term memory and RAG. Experimented with several tasks like memory read/write, conversational QA, multi-document QA.

EDUCATION

PhD (PMRF) in Computer Science and Engineering 2020-2025
Indian Institute of Technology, Kharagpur.
Under the guidance of **Dr. Saptarshi Ghosh**. (Coursework CGPA: **9.3**)
Thesis title: *Effects of COVID-19 pandemic on Vaccine-Opinions through Social Media Analyses*

MTech Degree in Computer Science and Engineering 2018-2020
Indian Institute of Technology, Kharagpur
Graduated with CGPA: **9.6** (Department Rank: **2**)
Thesis Title: *Summarizing Legal Case Documents: Incorporating Domain Knowledge in Summarization Algorithms*

BTech Degree in Computer Science and Technology 2014-2018
Indian Institute of Engineering Science and Technology, Shibpur
Graduated with CGPA: **9.1**

SKILLS

Human Languages	English (fluent), Bengali (native), Hindi
Computer Languages	Python (extensive experience), C, C++, Java, HTML, CSS
Tools/Technologies	LangGraph, Transformers, Huggingface, PyTorch, Flask, Docker, GCP Github Copilot, Git, Linux, LaTeX
Hobbies	Bass Guitar, Motorcycling, Swimming, Gaming, watch Formula **1** & STEM documentaries

AWARDS

- Awarded **PMRF fellowship** by Ministry of Education, Govt. of India. December 2020 cycle.
- **Best Student Paper** Award at ICAIL 2021
- Received several **Travel Grants** to attend SIGIR 2022 (Madrid), ICWSM 2024 (Buffalo), NAACL 2025 (Albuquerque).
- Qualified for the onsite **regionals of ACM ICPC** in **Dec 2016** (IIT Kharagpur region) and **Dec 2017** (Kolkata region).
- Winner and finalist of several **coding and robotics challenges** in different tech fests, e.g Overnite, Code Rush, Conquest, Embedtronix at Kshitij'16/17 (IIT Kgp), Ode to Code, Classified, Escapist, Konnect4 at Instruo'16/17(IIEST Shibpur)
- Winner of **Music Cup** of the Inter IIT Cultural meet **2019** (at IIT Bombay) and **2023** (at IIT Kharagpur).

PUBLICATIONS

- Rohit Dutta, Paramita Koley, Soham Poddar, Janardan Misra, Sanjay Podder, Navveen Balani Niloy Ganguly, Saptarshi Ghosh. **"Benchmarking the Energy Savings with Speculative Decoding Strategies"** *In Findings of the European Chapter of the Association for Computational Linguistics: EACL 2026*

• Soham Poddar, Paramita Koley, Janardan Misra, Niloy Ganguly, Saptarshi Ghosh. **“Brevity is the soul of sustainability: Characterizing LLM response lengths”** *In Findings of the Association for Computational Linguistics: ACL 2025*

• Soham Poddar, Paramita Koley, Janardan Misra, Niloy Ganguly, Saptarshi Ghosh. **“Towards Sustainable NLP: Insights from Benchmarking Inference Energy in Large Language Models”** *In Proceedings of Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL), 2025.*

• Soham Poddar, Subhendu Khatuya, Rajdeep Mukherjee, Niloy Ganguly, Saptarshi Ghosh. **“How COVID-19 has Impacted the Anti-Vaccine Discourse: A Large-Scale Twitter Study Spanning Pre-COVID and Post-COVID Era”** *In Proceedings of the 18th International AAAI Conference of Web and Social Media (ICWSM) 2024.*

• Soham Poddar, Rajdeep Mukherjee, Azlaan Mustafa Samad, Niloy Ganguly, Saptarshi Ghosh. **”MuLX-QA: Classifying Multi-Labels and Extracting Rationale Spans in Social Media Posts”** *ACM Transactions on the Web (TWEB), 2024.*

• Soham Poddar, Azlaan Mustafa Samad, Rajdeep Mukherjee, Niloy Ganguly, Saptarshi Ghosh. **“CAVES: A dataset to facilitate explainable classification and summarization of concerns towards COVID vaccines”**. *In Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR) 2022.*

• Soham Poddar, Mainack Mondal, Janardan Misra, Niloy Ganguly, and Saptarshi Ghosh. **“Winds of Change: Impact of COVID-19 on Vaccine-related Opinions of Twitter users”** *In Proceedings of the 16th International AAAI Conference of Web and Social Media (ICWSM) 2022.*

• Abhay Shukla, Paheli Bhattacharya, Soham Poddar, Rajdeep Mukherjee, Kripabandhu Ghosh, Pawan Goyal and Saptarshi Ghosh. **“Legal Case Document Summarization: Extractive and Abstractive Methods and their Evaluation”**. *The 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (AACL-IJCNLP), 2022.*

• Paheli Bhattacharya, Soham Poddar, Koustav Rudra, Kripabandhu Ghosh, and Saptarshi Ghosh. **“Incorporating Domain Knowledge for Extractive Summarization of Legal Case Documents”** *In Proceedings of the 18th International Conference on Artificial Intelligence and Law (ICAIL) 2021.*

• Soham Poddar, Biswajit Paul, Moumita Basu, and Saptarshi Ghosh. **“ICPR 2024 Competition on Multilingual Claim-Span Identification”**. *In Proceedings of the 27th International Conference on Pattern Recognition (ICPR), 2024.*

• Rahul Pullanikkat, Soham Poddar, Anik Das, Tushar Jaiswal, Vivek Kumar Singh, Moumita Basu, and Saptarshi Ghosh. **“Utilizing the Twitter social media to identify transportation-related grievances in Indian cities”**. *Social Network Analysis and Mining (SNAM), 2024.*

• Soham Poddar, Mainack Mondal, and Saptarshi Ghosh. **“A Survey on Disaster: Understanding the After-effects of Super-cyclone Amphan and Helping Hand of Social Media.”** *Advances in Urban Design and Engineering, Springer, 2022.*

• Ashish Kumar Layek, Soham Poddar, and Sekhar Mandal. **“Detection of Flood Images Posted on Online Social Media for Disaster Response.”** *In Proceedings of the 2nd International Conference on Advanced Computational and Communication Paradigms (ICACCP) 2019.*

SYSTEMS DEVELOPED

• **MESSAGE CHECK** is a fact-checking dashboard, and a predictive learning platform to identify existing fact-checks from www.vishvasnews.com that match dis/misinformation claims going viral. It also enables fact-checkers to predict misinformation around events and identify seasonal or event-based trends that cause a surge in misinformation. Given a multilingual query, it was processed using BM25+ retriever, a supervised Fasttext classifier and a small BERTScore model. It was then merged using Reciprocal Rank Fusion method to efficiently and effectively match debunked claims from a database.
Link to website: https://mdp.vishvasnews.com/

SELECTED BROAD PROJECTS

Analysis of Energy Efficiency of LLM Inference 2024-2025
IIT Kharagpur

• Benchmarked the energy usage of different LLMs for various NLP tasks under different scenarios, and the effect of different optimizations methodologies [e.g. I/O compression, model compression, speculative decoding] in reducing energy consumption.
• Showed that LLMs generate very long answers for factual queries, and formally categorize the information into different classes [e.g. minimal answer, additional information, reasoning, redundant information]. We highlight the trade-offs of such long answers that improve the user experience and utility but come at the expense of much higher energy consumption.
• We also explored some strategies to control the length and content composition of generated outputs, which can be used depending on the specific use case/user preferences; e.g. supervised fine-tuning, prompt-engineering with target length prediction, next token entropy mapping.

Characterizing User Opinions towards Vaccines on Twitter (PhD Thesis Work) 2020-2025
IIT Kharagpur

• We used automated NLP methods to systematically analyse and derive insights from the vaccine-opinions of various Twitter user-groups, and how these have evolved due to the COVID-19 pandemic. Beyond high-level Anti- and Pro-vax categorization, we tried to understand specific reasons why people are hesitant to take vaccines and what causes them to change their opinions. We then derive insights to help policy-makers address vaccine concerns, especially during future epidemics.

• Developed the large-scale annotated CAVES dataset for explainable multi-label classification of 12 vaccine concerns (e.g. side-effect, ineffectiveness, political, conspiracies). We also developed Transformer models (encoder-only, LLMs) and trained (SGD, SFT, LORA) them to identify concerns towards vaccines effectively, along with respective text span explanations.
• Explored prompt-tuning and fine-tuning LLMs to provide personalised arguments to counter anti-vaccine content on social media to resolve misconceptions, using a two-stage pipeline for identifying concerns followed by generating counters.

Summarizing Legal Case Documents (MTech Project and beyond) 2019-2022
IIT Kharagpur
• Compared performance of various summarization algorithms [generic and domain-specific; abstractive and extractive; classical ML vs Transformers based methods] on our curated set of UK and Indian Supreme courts' judgement documents.
• Created an unsupervised Linear Programming based method to systematically incorporate guidelines from legal experts into to create summaries with appropriate proportions of different rhetorical roles [e.g. fact, statutes, precedents, final judgement].

OFFICIALLY REVIEWED PAPERS FOR

• ACL Rolling Review (ACL ARR), 2025.

• Connection Science (CCOS) Journal, 2024.

• Artificial Intelligence and Law (ARTI) Journal, 2024.

• The 18th International AAAI Conference of Web and Social Media (ICWSM) 2024

• Online Social Networks and Media (OSNEM) Journal, 2023.

• The 16th International AAAI Conference of Web and Social Media (ICWSM) 2022

SHARED TASKS/COMPETITIONS ORGANIZED

• Multilingual Claim Span Identification at ICPR 2024. https://sites.google.com/view/icpr24-csi/home

• Artificial Intelligence on Social Media at FIRE 2023. https://sites.google.com/view/aisome/aisome

• Information Retrieval from Microblogs during Disasters at FIRE 2022. https://sites.google.com/view/irmidis-fire2022/irmidis

• Information Retrieval from Microblogs during Disasters at FIRE 2021. https://sites.google.com/view/irmidisfire2021

REFERENCES

1. **Prof. Niloy Ganguly**, Professor/HOD
   Dept. of Computer Science and Engineering, Indian Institute of Technology, Kharagpur.
   Email: *head@cse.iitkgp.ac.in*
2. **Prof. Saptarshi Ghosh**, Associate Professor, PhD Supervisor
   Dept. of Computer Science and Engineering, Indian Institute of Technology, Kharagpur.
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